a galaxy is, the faster it moves away from that point. Note that no one penny is singled out as special, also in keeping with our belief that no galaxy in the universe is special or the center of the expansion of space.By attributing the observed motion of galaxies to the swelling of space, general relativity provides an explanation that not only treats all locations in space symmetrically, but also accounts for all of Hubble's data in one fell swoop. It is this kind of explanation, one that elegantly steps outside the box (in this case, one that actually *uses* the "box"—space, that is) to explain observations with quantitative precision and artful symmetry, that physicists describe as almost being too beautiful to be wrong. There is essentially universal agreement that the fabric of the space is stretching.

Time in an Expanding Universe

Using a slight variation on the balloon model, we can now understand more precisely how symmetry in space, even though space is expanding, yields a notion of time that applies uniformly across the cosmos. Imagine replacing each penny by an identical clock, as in Figure 8.3. We know from relativity that identical clocks will tick off time at different rates if they are subject to different physical influences—different motions, or different gravitational fields. But the simple yet key observation is that the complete symmetry among all Lincolns on the inflating balloon translates to complete symmetry among all the clocks. All the clocks experience identical physical conditions, so all tick at exactly the same rate and record identical amounts of elapsed time. Similarly, in an expanding universe in which there is a high degree of symmetry among all the galaxies, *clocks that move along with one or another galaxy must also tick at the same rate and hence record an identical amount of elapsed time.* How could it be otherwise? Each clock is on a par with every other, having experienced, on average, nearly identical physical conditions. This again shows the stunning power of symmetry. Without any calculation or detailed analysis, we realize that the uniformity of the physical environment, as evidenced by the uniformity of the microwave background radiation and the uniform distribution of galaxies throughout space,[8] allows us to infer uniformity of time.